# Virtual-Structure Formation Control of a Ground Robot and a Quadrotor with Null-Space-Based Obstacle Avoidance

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***Abstract*—This report describes the design and experimental validation of a formation controller for a heterogeneous team composed of a differential-drive ground robot (AgileX LIMO) and a quadrotor (Parrot Bebop 2), based on the virtual-structure paradigm. The controller follows an inner–outer loop architecture: an outer kinematic loop drives the formation variables along a Bernoulli-lemniscate trajectory while keeping the drone above the ground robot, and an inner loop performs dynamic compensation for each robot using identified dynamic models. Obstacle avoidance is embedded as the highest-priority subtask through a null-space-based behavioral (NSB) scheme, with a Gaussian potential field describing the obstacle. Experiments were carried out in the LAB-AIR arena using OptiTrack motion capture and ROS. The methodology and equations follow the textbook by Sarcinelli-Filho and Carelli [1].**

***Index Terms*—formation control, virtual structure, null-space, potential field, quadrotor, differential-drive robot**

## I. Introduction

The goal of this work is to control a two-robot formation formed by a ground differential-drive robot (LIMO, denoted L1) and a quadrotor (Bebop 2, denoted B1) so that the formation tracks a desired trajectory in the plane while the quadrotor is kept at a constant altitude above the ground robot. A static cylindrical obstacle is placed in the workspace and must be avoided with priority over the formation task.

The controller is designed according to the virtual-structure paradigm and the null-space-based behavioral control approach described in [1]. The complete control law is of the inner–outer loop type: the outer loop is a kinematic formation controller expressed in the cluster space, and the inner loop is a dynamic compensator, one for each robot, mapping desired velocities into the commands actually sent to the robots. This report presents the model, the control equations, the experimental parameters, and the obtained results, including an intermediate step in which only the ground robot is commanded.

## II. Formation Modeling

### A. Cluster variables and forward map

The point of interest of the formation is the control point of the LIMO, displaced a distance $a$ from its center of gravity along a line at zero degrees with the robot $X$ axis. The formation (cluster) variables are

$$\mathbf{q} = \begin{bmatrix} x_f & y_f & z_f & \rho_f & \alpha_f & \beta_f \end{bmatrix}^T, \tag{1}$$

where $(x_f, y_f, z_f)$ is the position of the control point, $\rho_f$ is the distance to the drone, $\alpha_f$ is the azimuth and $\beta_f$ the elevation of the segment linking the control point to the drone. The forward map from cluster variables to robot positions, following [1], is

$$x_1 = x_f, \quad y_1 = y_f, \quad z_1 = z_f = 0, \tag{2}$$
$$x_2 = x_f + \rho_f \cos\beta_f \cos\alpha_f, \tag{3}$$
$$y_2 = y_f + \rho_f \cos\beta_f \sin\alpha_f, \tag{4}$$
$$z_2 = z_f + \rho_f \sin\beta_f, \tag{5}$$

where subscripts 1 and 2 denote the LIMO and the drone, respectively.

### B. Desired formation and trajectory

The desired formation keeps the drone directly above the control point, i.e. $\alpha_f = 0$, $\beta_f = 90°$ and $\rho_f = 1.5$ m, so that $x_2 = x_f$, $y_2 = y_f$ and $z_2 = 1.5$ m. The desired trajectory is a Bernoulli-lemniscate in the plane,

$$x_d = 0.75\,\sin\!\left(\tfrac{2\pi t}{40}\right), \tag{6}$$
$$y_d = 0.75\,\sin\!\left(\tfrac{4\pi t}{40}\right), \tag{7}$$

with $z_f = 0$ and the drone at constant altitude of $1.5$ m. The control loop runs at 30 Hz, i.e. the sampling period is $T = 1/30$ s.

*Remark on the singularity.* With $\beta_f = 90°$ the drone lies exactly above the control point and the azimuth $\alpha_f$ becomes undefined, which is a singular configuration of the full formation Jacobian [1]. For this reason the drone is not commanded through the inverse of the full $6 \times 6$ Jacobian; instead, it follows the planar motion of the control point and its own altitude controller, which is equivalent to the desired formation but avoids the singularity.

## III. Control Design

### A. Outer loop: kinematic formation controller

For each planar coordinate of the control point, the outer loop uses a feedforward plus saturated proportional law, in the form of the cluster-space controller of [1],

$$v_x = \dot{x}_d + L_x \tanh\left(\tfrac{k_x}{L_x}\,\tilde{x}\right), \qquad (8)$$

$$v_y = \dot{y}_d + L_y \tanh\left(\tfrac{k_y}{L_y}\,\tilde{y}\right), \qquad (9)$$

$$v_z = k_z\,(z_{2d} - z_2), \qquad (10)$$

where $\tilde{x} = x_d - x_f$, $\tilde{y} = y_d - y_f$, and $\dot{x}_d, \dot{y}_d$ are the analytical derivatives of (6)–(7). The pair $(v_x, v_y)$ is the desired velocity of the control point expressed in the global frame.

### B. Null-space-based obstacle avoidance with a Gaussian potential field

Obstacle avoidance is embedded as the highest-priority subtask through the null-space-based behavioral control approach [1]. The obstacle, a cylinder centered at $(x_{obs}, y_{obs})$, is described by the Gaussian potential field

$$V = \exp\Big[-\left(\tfrac{x-x_{obs}}{a_p}\right)^n - \left(\tfrac{y-y_{obs}}{b_p}\right)^n\Big], \qquad (11)$$

with $n$ a positive even integer and $a_p, b_p$ tuning the spread. The associated Jacobian is the gradient of the potential,

$$\mathbf{J}_o = \begin{bmatrix} \dfrac{\partial V}{\partial x} & \dfrac{\partial V}{\partial y} \end{bmatrix}, \qquad (12)$$

and the avoidance subtask drives the potential down to a small desired value $V_d$ using

$$\dot{\mathbf{x}}_{avoid} = \mathbf{J}_o^{\dagger}\left(\dot{V}_d + k_{obs}\,(V_d - V)\right), \quad \dot{V}_d = 0. \qquad (13)$$

The null-space projector of the obstacle task, $\mathbf{N} = \mathbf{I} - \mathbf{J}_o^{\dagger}\mathbf{J}_o$, is then used to *unite* the two tasks, so that the formation-tracking velocity is projected into the null space of the avoidance task,

$$\dot{\mathbf{x}} = \dot{\mathbf{x}}_{avoid} + \mathbf{N}\begin{bmatrix} v_x \\ v_y \end{bmatrix}. \qquad (14)$$

The avoidance is activated only when the control point is inside the circular influence zone of the obstacle, i.e. when $d <$ influence, where $d$ is the distance from the control point to the obstacle center. Outside the influence zone the formation task acts alone.

### C. Inner loop: dynamic compensation of the LIMO

The desired velocity of the control point is first mapped into desired linear/angular velocities of the LIMO through the inverse kinematics of the control point,

$$v_d = \cos\psi\, v_x + \sin\psi\, v_y, \qquad (15)$$

$$\omega_d = \tfrac{1}{a}\left(-\sin\psi\, v_x + \cos\psi\, v_y\right), \qquad (16)$$

where $\psi$ is the LIMO heading. The dynamic compensator then produces the reference velocities using the identified model of [1],

$$v = \theta_1\sigma_v - \theta_3\omega_d\,\omega + \theta_4 v_d, \qquad (17)$$

$$\omega = \theta_2\sigma_\omega + \theta_3\omega v_d + \theta_6\omega_d + \theta_5 v\,\omega_d - \theta_3\omega_d v, \qquad (18)$$

with $\sigma_v = \dot{v}_d + l_u \tanh(\frac{k_u}{l_u}(v_d - v))$ and $\sigma_\omega = \dot{\omega}_d + l_\omega \tanh(\frac{k_\omega}{l_\omega}(\omega_d - \omega))$, where $v, \omega$ are estimated from the OptiTrack pose. The identified parameters are listed in Table I.

### D. Inner loop: dynamic compensation of the quadrotor

The quadrotor dynamic model, written in local (body) variables, is $\dot{\mathbf{v}} = \mathbf{f}_1\mathbf{u} - \mathbf{f}_2\mathbf{v}$. The inner loop inverts this model,

$$\mathbf{u} = \mathbf{f}_1^{-1}\left(\dot{\mathbf{v}} + \mathbf{f}_2\mathbf{v}\right), \qquad (19)$$

where the desired body velocity is obtained by rotating the global command by the drone yaw $\psi_b$. The horizontal command combines the control-point velocity with the drone's own position error toward its target $(x_{2d}, y_{2d})$, which was found necessary in practice (Section V). A first-order low-pass filter is applied to the desired velocity before numerical differentiation to attenuate measurement noise in $\dot{\mathbf{v}}$.

### E. Saturation and safety

Soft ($\tanh$) and hard saturation limits are applied to all commands. Safety routines were implemented as required for the flight tests: a joystick emergency-stop button that zeros the commands and lands the drone; a `try/catch` guard that lands the drone on any run-time error; a virtual wall ($|x| > 2$, $|y| > 2$, $z > 1.8$ m); an OptiTrack body-loss watchdog ($>$ 0.5 s); and a pre-flight test that reads the pose and checks the control signal without taking off.

TABLE I
EXPERIMENTAL PARAMETERS

| Symbol | Description | Value |
|---|---|---|
| $T$ | sampling period | $1/30$ s |
| $t_f$ | experiment duration | 120 s |
| $a$ | control-point offset | 0.10 m |
| $\rho_f$ | formation distance | 1.5 m |
| $\alpha_f, \beta_f$ | azimuth, elevation | 0°, 90° |
| $k_x, k_y$ | outer-loop gains | 1.0, 1.0 |
| $L_x, L_y$ | outer-loop saturation | 0.8, 0.8 |
| $k_z$ | altitude gain | 1.0 |
| $k_u, k_\omega, l_u, l_\omega$ | LIMO inner-loop | $4, 4, 1, 1$ |
| $\theta_{1\ldots6}$ | LIMO model | 0.152, 0.095, 0.003, 0.984, −0.045, 1.642 |
| $x_{obs}, y_{obs}$ | obstacle center | (−0.2, 0.425) m |
| $r_{obs}$ | obstacle radius | 0.15 m |
| influence | influence-zone radius | 0.25 m |
| $a_p, b_p, n$ | potential field | 0.30, 0.30, 4 |
| $k_{obs}, V_d$ | avoidance gain, target | 0.5, 0.1 |
| $v_{max}, \omega_{max}, v_{z,max}$ | saturation limits | 1, 1, 0.4 |

## IV. Experimental Setup

The experiment was carried out in the LAB-AIR arena. Robot poses were obtained from an OptiTrack motion-capture system published over ROS, and velocity commands were sent to each robot over ROS as well. The control loop ran on MATLAB at 30 Hz. The LIMO started at $(0.4, -0.25, 0)$ m aligned with the global $X$ axis, and the drone started about 0.30 m to its left, at $(0.4, 0.05)$ m. The obstacle was a bucket placed at $(-0.2, 0.425, 0)$ m.

# V. Results

## A. Intermediate step: ground robot only

As an intermediate validation step, the controller was run with the drone disabled, commanding only the LIMO along the lemniscate with obstacle avoidance active. Figure 1 shows the desired trajectory, the LIMO path, and the obstacle with its influence zone. The control point tracks the lemniscate and deviates around the obstacle when inside the influence zone, returning to the reference afterwards.

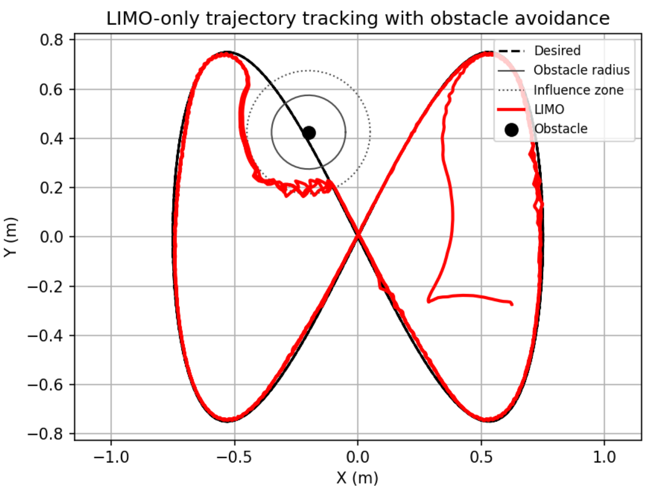


Fig. 1. LIMO-only intermediate test: trajectory tracking with obstacle avoidance.

For this run, the control-point tracking error (distance between the control point and the desired trajectory) had a mean of $0.068$ m and an RMS of $0.117$ m over the whole experiment; after the initial transient (first $10$ s) the mean error dropped to $0.056$ m with an RMS of $0.082$ m.

## B. Full formation

Figure 2 shows the complete experiment, with the LIMO and the drone tracking the lemniscate while the drone remains above the control point, and the formation deviating around the obstacle.

The altitude of the drone was kept close to the $1.5$ m reference, with a mean of $1.489$ m and a standard deviation of $0.056$ m over the run ($1.495 \pm 0.039$ m after the initial transient), as shown in Fig. 3. The horizontal formation error (distance between the drone and the control point in the plane) had a mean of $0.195$ m, reflecting the larger transients of the aerial vehicle, as shown in Fig. 4. The minimum distance from the control point to the obstacle center along the whole experiment was $0.205$ m, which is above the obstacle radius of $0.15$ m; the control point spent 125 samples ($\approx 4.2$ s) inside the influence zone, during which the avoidance subtask was active (Fig. 5).

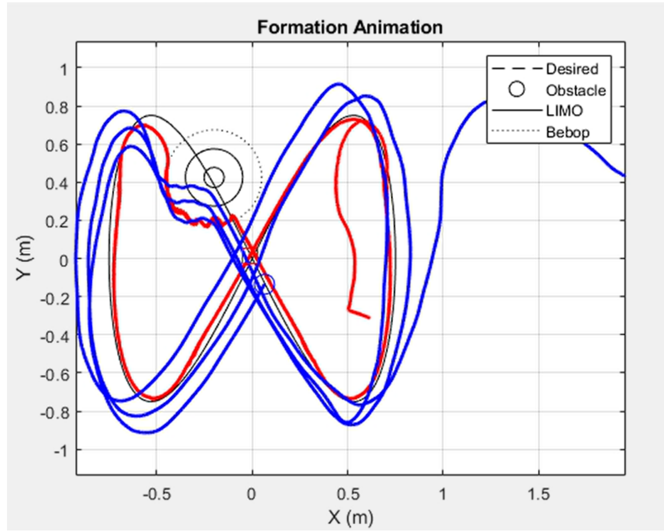


Fig. 2. Full formation experiment: desired trajectory, LIMO path, drone path, and obstacle.

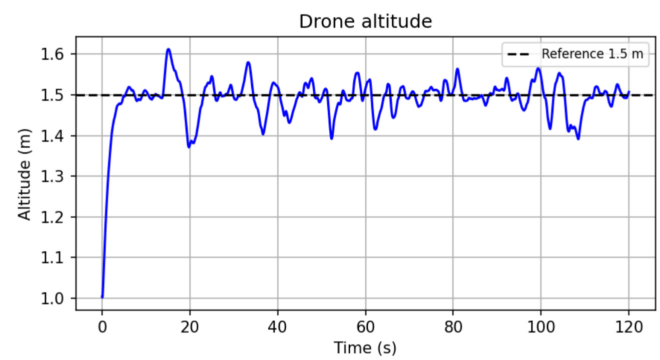


Fig. 3. Drone altitude during the full experiment.

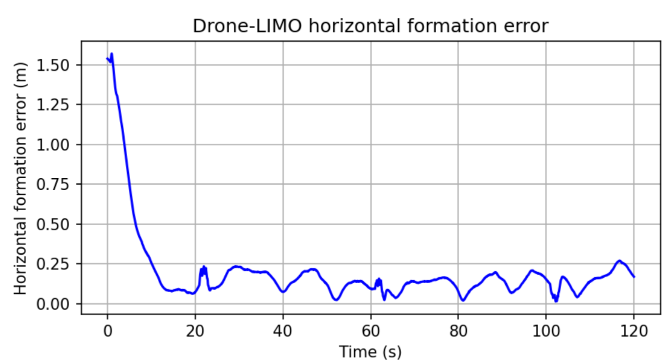


Fig. 4. Horizontal formation error (drone with respect to the control point).

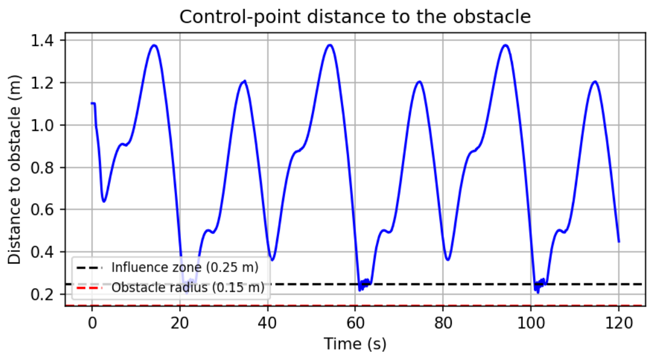


Fig. 5. Distance from the control point to the obstacle, with the influence zone and obstacle radius indicated.

### *C. Discussion*

The intermediate LIMO-only test confirmed that the outer kinematic loop plus the dynamic compensator track the lemniscate accurately and that the Gaussian potential field keeps the control point outside the obstacle radius. In the full formation, the ground robot tracking remained accurate, while the drone presented larger horizontal excursions. This was traced to the horizontal command of the drone: in an early version, the drone reused the control-point velocity without feedback of its own horizontal position, so only the altitude channel (which uses the drone's own error) behaved well. Adding the drone's own position-error feedback in (19), together with a low-pass filter before the numerical derivative of the desired velocity, substantially reduced the horizontal oscillation. The residual horizontal error is consistent with the higher sensitivity of the aerial vehicle to motion-capture noise and to the derivative term of the inner loop.

## VI. Conclusion

A virtual-structure formation controller with inner-outer loop architecture was designed and tested for a heterogeneous LIMO–Bebop team. Obstacle avoidance was embedded as the highest-priority subtask through a null-space-based scheme using a Gaussian potential field, with the formation task projected into the null space of the avoidance task. Experiments in the LAB-AIR arena, first with the ground robot alone and then with the full formation, showed accurate trajectory tracking, altitude regulation near $1.5$ m, and obstacle clearance above the obstacle radius. The main practical difficulty was the horizontal stabilization of the quadrotor, addressed by adding position feedback and filtering in the inner loop.

## Appendix

1) *Videos of the experiment are available by clicking here*
2) *The complete MATLAB/ROS source code used in this work is available by clicking here*

## References

[1] M. Sarcinelli-Filho and R. Carelli, *Control of Ground and Aerial Robots*. Springer.